

Class 11: Structural Bioinformatics pt2.

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Background

We saw last day that the PDB has 209,886 entries (Oct/Nov 2025). UniprotKB (i.e. protein sequence database) has 199,579,901 entries.

```
209886/199579901 * 100
```

```
[1] 0.1051639
```

So the PDB has only 0.1% coverage of the main sequence database.

Enter AlphaFold data base (AFDB). < <https://alphafold.ebi.ac.uk/> > that attempts to provide computed models for all sequences in UniProt.

AlphaFold

AlphaFold has three main outputs

- the predicted coordinates (PDB file)
- A local quality score called **pLDDT** (one for each amino acid)
- a second quality score **PAE** predicted aligned error (for each pair of amino-acid)

We can run AlphaFold ourselves if we are not happy with AFDB (i.e. no coverage or poor model).

Interpreting/analyzing AF results in R

```
results_dir <- "HIVPR_dimer_23119.result/HIVPR_dimer_23119/"
```

```
pdb_files <- list.files(path=results_dir, pattern="*.pdb", full.names = TRUE)
```

```
library(bio3d)
```

Warning: package 'bio3d' was built under R version 4.3.3

```
library(bio3dview)
```

```
pdbbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_msa.pdb
HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_msa.pdb
HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_msa.pdb
HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_msa.pdb
HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_msa.pdb
.....
```

Extracting sequences

```
pdb/seq: 1   name: HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_msa.pdb
pdb/seq: 2   name: HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_msa.pdb
pdb/seq: 3   name: HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_msa.pdb
pdb/seq: 4   name: HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_msa.pdb
pdb/seq: 5   name: HIVPR_dimer_23119.result/HIVPR_dimer_23119/HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_msa.pdb
```

```
rd <- rmsd(pdbbs, fit=T)
```

Warning in rmsd(pdbbs, fit = T): No indices provided, using the 198 non NA positions

```
range(rd)
```

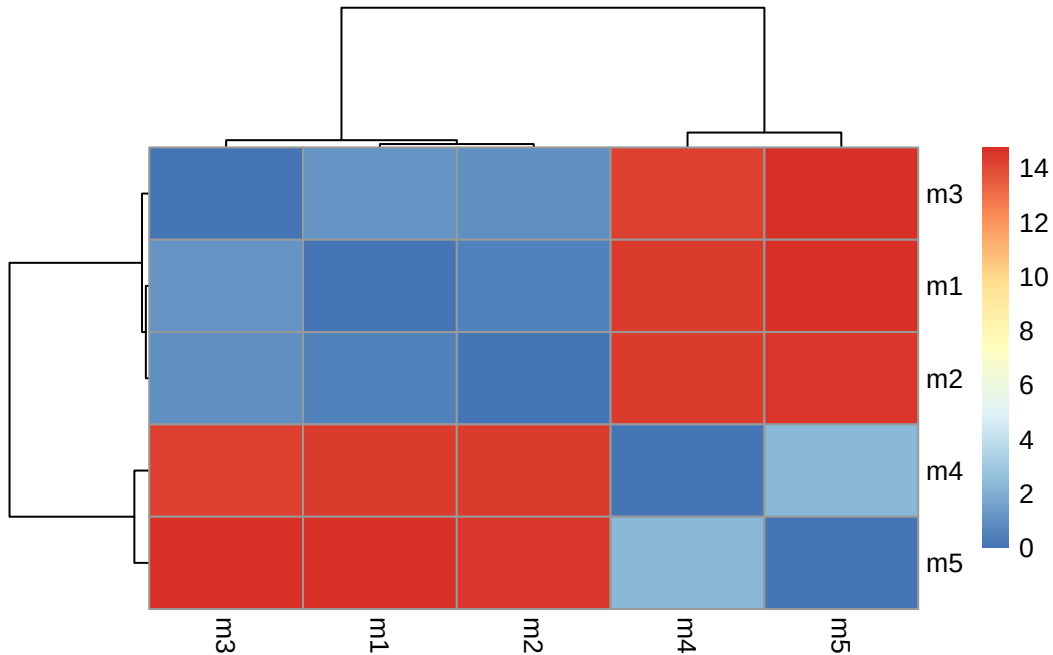
```
[1] 0.000 14.754
```

Analysis

```
library(pheatmap)
```

Warning: package 'pheatmap' was built under R version 4.3.3

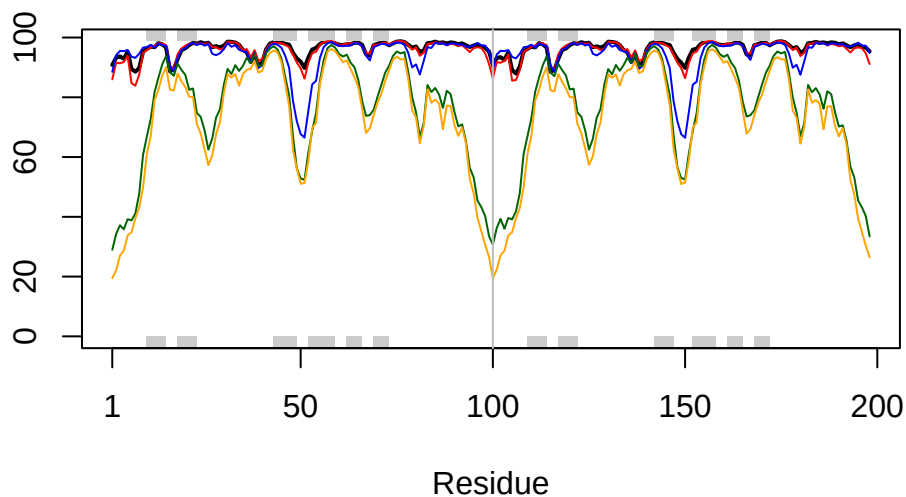
```
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ="l", lwd=2, sse=pdb)
points(pdb$b[2,], typ="l", col="red")
points(pdb$b[3,], typ="l", col="blue")
points(pdb$b[4,], typ="l", col="darkgreen")
points(pdb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



```
core <- core.find(pdb)
```

```
core size 197 of 198 vol = 9836.196
core size 196 of 198 vol = 6800.793
core size 195 of 198 vol = 1322.091
core size 194 of 198 vol = 1032.249
core size 193 of 198 vol = 944.133
core size 192 of 198 vol = 894.048
core size 191 of 198 vol = 829.724
core size 190 of 198 vol = 766.628
core size 189 of 198 vol = 728.681
core size 188 of 198 vol = 693.317
core size 187 of 198 vol = 656.166
core size 186 of 198 vol = 622.063
core size 185 of 198 vol = 586.601
core size 184 of 198 vol = 565.358
core size 183 of 198 vol = 532.747
core size 182 of 198 vol = 511.421
core size 181 of 198 vol = 489.077
core size 180 of 198 vol = 468.599
core size 179 of 198 vol = 449.222
core size 178 of 198 vol = 433.122
```

core size 177 of 198	vol = 418.651
core size 176 of 198	vol = 405.069
core size 175 of 198	vol = 391.855
core size 174 of 198	vol = 381.006
core size 173 of 198	vol = 371.54
core size 172 of 198	vol = 355.539
core size 171 of 198	vol = 345.156
core size 170 of 198	vol = 335.982
core size 169 of 198	vol = 325.275
core size 168 of 198	vol = 313.603
core size 167 of 198	vol = 302.908
core size 166 of 198	vol = 293.343
core size 165 of 198	vol = 284.547
core size 164 of 198	vol = 277.932
core size 163 of 198	vol = 269.592
core size 162 of 198	vol = 258.167
core size 161 of 198	vol = 246.856
core size 160 of 198	vol = 239.073
core size 159 of 198	vol = 234.174
core size 158 of 198	vol = 226.309
core size 157 of 198	vol = 221.212
core size 156 of 198	vol = 214.951
core size 155 of 198	vol = 206.027
core size 154 of 198	vol = 197.301
core size 153 of 198	vol = 190.889
core size 152 of 198	vol = 185.403
core size 151 of 198	vol = 179.154
core size 150 of 198	vol = 174.709
core size 149 of 198	vol = 167.269
core size 148 of 198	vol = 160.866
core size 147 of 198	vol = 156.173
core size 146 of 198	vol = 148.395
core size 145 of 198	vol = 143.389
core size 144 of 198	vol = 138.428
core size 143 of 198	vol = 132.99
core size 142 of 198	vol = 126.924
core size 141 of 198	vol = 121.257
core size 140 of 198	vol = 116.407
core size 139 of 198	vol = 112.241
core size 138 of 198	vol = 107.813
core size 137 of 198	vol = 104.807
core size 136 of 198	vol = 100.968
core size 135 of 198	vol = 97.21

core size 134 of 198	vol = 92.54
core size 133 of 198	vol = 87.964
core size 132 of 198	vol = 85.621
core size 131 of 198	vol = 81.514
core size 130 of 198	vol = 77.627
core size 129 of 198	vol = 74.881
core size 128 of 198	vol = 72.642
core size 127 of 198	vol = 70.352
core size 126 of 198	vol = 68.627
core size 125 of 198	vol = 66.349
core size 124 of 198	vol = 64.019
core size 123 of 198	vol = 61.763
core size 122 of 198	vol = 58.758
core size 121 of 198	vol = 56.354
core size 120 of 198	vol = 54.129
core size 119 of 198	vol = 51.611
core size 118 of 198	vol = 49.502
core size 117 of 198	vol = 48.045
core size 116 of 198	vol = 46.489
core size 115 of 198	vol = 44.599
core size 114 of 198	vol = 43.185
core size 113 of 198	vol = 40.978
core size 112 of 198	vol = 39.029
core size 111 of 198	vol = 36.363
core size 110 of 198	vol = 33.953
core size 109 of 198	vol = 31.343
core size 108 of 198	vol = 29.304
core size 107 of 198	vol = 27.188
core size 106 of 198	vol = 25.655
core size 105 of 198	vol = 24.03
core size 104 of 198	vol = 22.504
core size 103 of 198	vol = 20.906
core size 102 of 198	vol = 19.803
core size 101 of 198	vol = 18.215
core size 100 of 198	vol = 15.652
core size 99 of 198	vol = 14.762
core size 98 of 198	vol = 11.574
core size 97 of 198	vol = 9.391
core size 96 of 198	vol = 7.327
core size 95 of 198	vol = 6.123
core size 94 of 198	vol = 5.592
core size 93 of 198	vol = 4.689
core size 92 of 198	vol = 3.66

```

core size 91 of 198  vol = 2.779
core size 90 of 198  vol = 2.147
core size 89 of 198  vol = 1.725
core size 88 of 198  vol = 1.158
core size 87 of 198  vol = 0.88
core size 86 of 198  vol = 0.682
core size 85 of 198  vol = 0.527
core size 84 of 198  vol = 0.37
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44

```

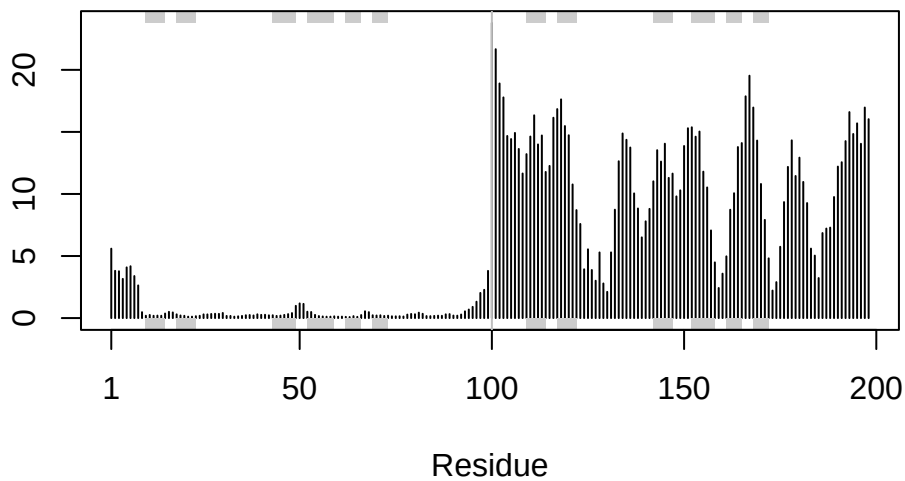
```
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
```

```
rf <- rmsf(xyz)
```

```

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")

```



Predicted Alignment Error

```
library(jsonlite)
```

Warning: package 'jsonlite' was built under R version 4.3.3

```
# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae2 <- read_json(pae_files[2],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)
```

```
attributes(pae2)
```

```
$names
[1] "plddt"    "max_pae" "pae"      "ptm"      "iptm"
```

```
head(pae2$plddt)
```

```
[1] 86.00 91.56 91.38 91.81 93.88 84.69
```

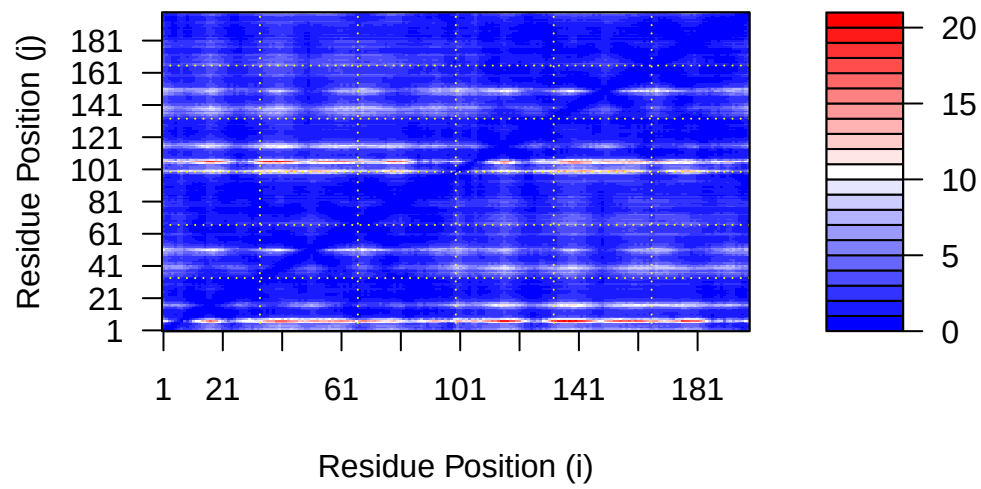
```
pae2$max_pae
```

```
[1] 20.04688
```

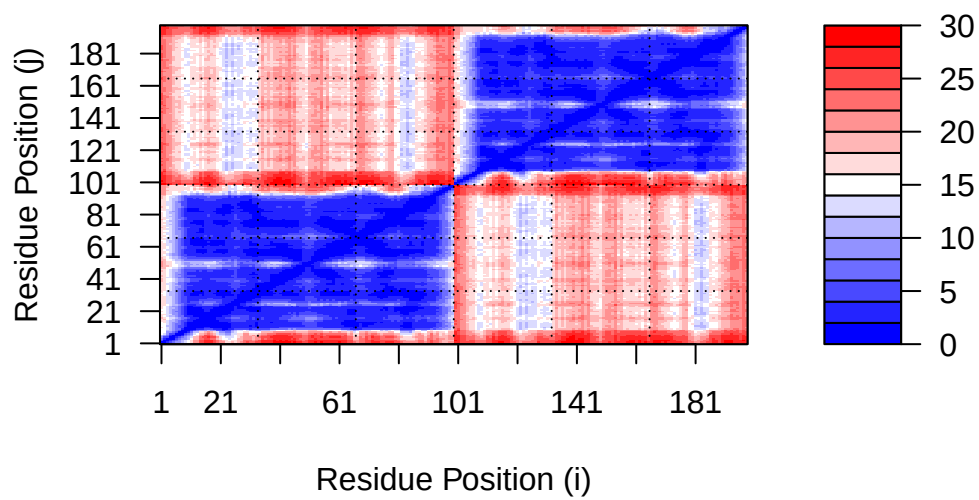
```
pae5$max_pae
```

```
[1] 29.59375
```

```
plot.dmat(pae2$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```

```
plot.dmat(pae5$paes,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



```
plot.dmat(pae2$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30)) #to plot on the same z range
```

